Topic C: proofs and their costs

Human Compression project

The common paper, *Changes of chamber generators and binary translation groups*, supports four complete applications: Janko geometries (C1), halving generalized cubes (C2), Petrials (C3), and equal-parameter polyhedra (C4). The fixed comparison preserves forty source clauses, including their explicit corrections, retained theorems and two excluded auxiliary assertions.

The complete written comparison. The common paper costs 27,131 characters and the four applications cost 44,146. The three available source PDFs contribute 60,256. If x is the unknown cost of the original L108 PDF, then

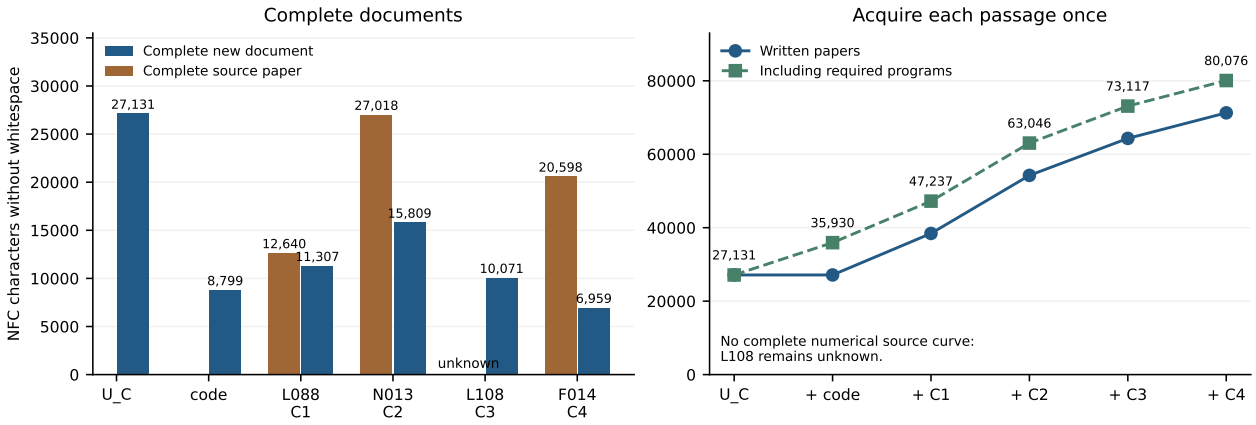
$$R_C^{\text{written}} = \frac{x + 60\,256}{71\,277}.$$

There is no numerical overall factor. The full 10,071-character Petrial application remains in the denominator. Its source browser rendering fixes the mathematical statement boundary, but is not a substitute PDF measurement.

Both complete GAP programs are required for the Janko application. Their printed appendix costs 8,799, charged once, giving the separate comparison

$$R_C^{\text{programs}} = \frac{x + 60\,256}{80\,076}.$$

Counting the two full raw files instead gives denominator 79,600. This is an alternative representation of the same programs, not an additional charge.



All figures on this page use NFC characters without whitespace in complete extracted PDF text, with the reviewed furniture exclusions and glyph calibration. The same accounting is supplied for all thirty policy coordinates. Unextracted graphics and expanded outside-proof costs remain unknown.

After acquiring the whole common theory and both programs. These are the admitted complete-source / complete-application factors.

Source / application	Source	Application	Conditional factor
L088 / C1	12,640	11,307	1.118
N013 / C2	27,018	15,809	1.709
L108 / C3	unknown	10,071	unknown
F014 / C4	20,598	6,959	2.960

A declared acquisition order. Acquire the complete U_C, the complete program appendix, and then C1, C2, C3 and C4. Every application includes its definitions, title, context, references and all proof passages. Let K_i be the union of passage owners paid by stage i. The new cost is $c(K_i \setminus K_{i-1})$; a prerequisite already in K_{i-1} is not charged again. No cost is shared across topics.

Added	New cost	Written total	With programs	Prior text used
U_C	27,131	27,131	27,131	—
CODE	8,799	27,131	35,930	—
C1	11,307	38,438	47,237	17,032
C2	15,809	54,247	63,046	16,913
C3	10,071	64,318	73,117	5,161
C4	6,959	71,277	80,076	22,407

The last column measures already acquired passages used by the complete local paper's recorded arguments. It is not an extra subtraction from that paper's cost. The historical cumulative written costs are 12,640, then 39,658, then $x + 39\,658$, and finally $x + 60\,256$. Thus no complete numerical historical acquisition curve is available. The sequence is a reading order, not an estimate of research effort or a shortest possible proof.

What a smaller proof route measures. The paired graph records prerequisite-to-use arrows for each of the forty clauses and for the four complete contracts. The additional identification of the equal-parameter construction with the earlier polygon construction stays separate. Each route's known cost is the union of its exclusive passage owners. It can omit context still paid in the complete paper. Accordingly these route subtotals are explanations, not replacement denominators or new compression factors. Parent results and their proof branches may share a statement owner; box costs overlap, while every total counts that passage once.

There are 156 graph nodes: 64 historical, 62 current, 26 retained external inputs and four verification components, with 249 typed edges. Their categories include definitions, statements and proof branches, at different levels of detail. No node-count ratio is defined. The accounting has 199 text owners and 195 selected-text spans. It also assigns excluded furniture to explicit owners for the untrimmed sensitivity; no printed text is silently dropped.

Scope and verification. The accepted logical revision restores the seed and subgroup assumptions of the historical Janko calculation without extending the string-only recognition theorem to nonstring exercise cases. It records the historical halving C-group premise and Cayley–Hamilton use, and keeps the current displacement proof independent of unnecessary binary-module assumptions. Geometry is established before naming the Janko residues. All source qualifications remain attached to their original clauses.

The original L108 PDF, outside-proof expansions and graphic costs remain unknown. The existing sixteen Lean declarations verify specified displacement, intersection and semidirect-product components; they are not full formalizations of the applications. The two complete Janko programs are preserved unchanged. The cost checker verifies full-document conservation, unions and deliberate corruptions. The offline HTML shows all proof routes, locations and unknown inputs; its deterministic DOM test is not a real-browser layout test. This new ownership and reader layer awaits independent review. It changes no mathematical contract, source or question credit, or admitted paper factor.